Can ERA5-Land Wind Stand In for LICRICE TC Wind?

Replicating Young & Hsiang, and swapping the exposure measure

September 8, 2026

Roadmap

1. **Setup** — the panel, and the specification we are matching.
2. **Replication** — does the Young–Hsiang headline reproduce? *Yes.*
3. **Comparing the two wind measures** — descriptive comparison of `maxs` and ERA5-Land wind.
4. **Results**
5. **Why the ERA5 coefficients are larger** — diagnostics: scaling, masking, season and state placebos, and leave-one-out.

Summary. The replication reproduces the published estimate. ERA5-Land wind is associated with mortality in the cyclone season, but does not measure the same events set as `maxs`.

Setup

Purpose

1. **Replicate** the Young–Hsiang (2024) main result: a 20-year distributed lag of tropical-cyclone wind on all-cause monthly mortality.
2. **Swap** their LICRICE wind measure (`maxs`) for the ERA5-Land wind we merged onto the same panel, and run the two against each other.

Panel. U.S. state × month, 36 states, 1950–2012 in estimation (lags reach back to 1930). Outcome is all-cause deaths per 100,000 per month.

Panel built by the merge documented in `2_analysis/wind_tc_validation/README.md`; original code at `Global-Policy-Lab/young_hsiang_tc_mortality`.

The specification

Young and Hsiang estimate, for state s in month-of-sample τ,

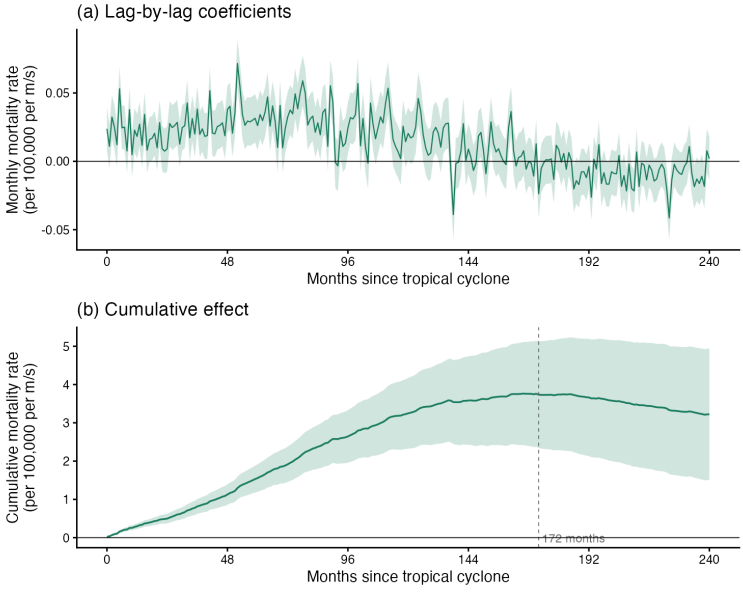
$$y_{s\tau} = \sum_{j=0}^{240} \beta_j W_{s,\tau-j} + \underbrace{\mu_{sm}}_{\substack{\text{state} \times \\ \text{cal. month}}} + \underbrace{\delta_\tau}_{\substack{\text{month-of-} \\ \text{sample}}} + \underbrace{f_s(\tau)}_{\substack{\text{state 8th-order} \\ \text{polynomial in } \tau}} + \underbrace{g_s(T_{s\tau})}_{\substack{\text{state quadratic} \\ \text{in temperature}}} + \varepsilon_{s\tau}$$

plus a state-by-calendar-month linear trend. The object of interest is the **cumulative** effect $B(h) = \sum_{j=0}^{h} \beta_j$; they report it at $h = 172$ months.

Their `areg` uses iid standard errors, so we match that (caveats on the replication slide). We estimate in `fixest`, absorbing the same column space (241 wind lags, ≈2,000 absorbed parameters, 27,216 observations).

Replication

Replication: the Young–Hsiang main result

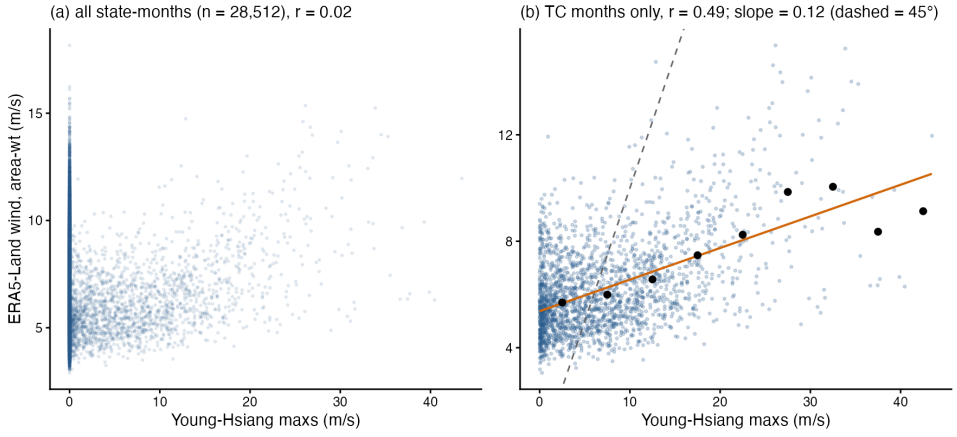


Replication: caveats

- `tavg` is missing for 2013–2015, so the temperature control truncates estimation at 2012. This is a property of Young and Hsiang's own data: their published `.dta` has the same gap, so their regressions truncate at 2012 too.
- With 36 states and 241 lag coefficients the state-clustered VCOV is rank-deficient (cluster rank ≤ 36), so clustered SEs are not reported here — Young and Hsiang use randomization inference rather than a cluster correction. We have not reproduced that step. **We use 'iid' SEs here.**

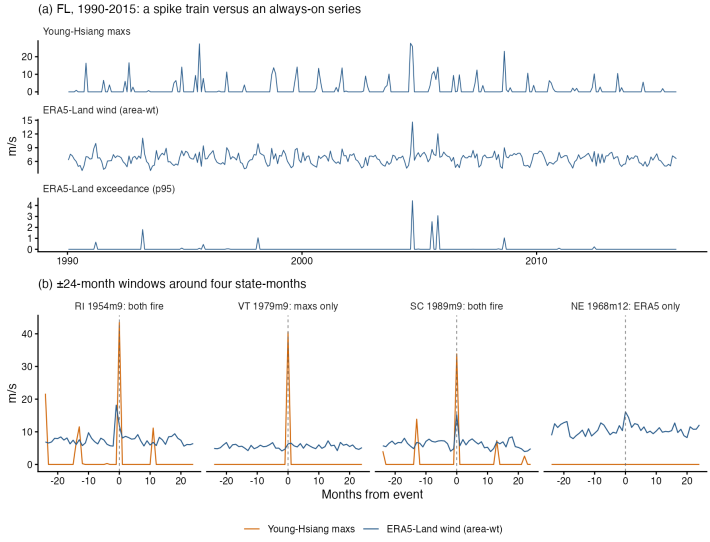
Comparing the two wind measures

ERA5-Land against `maxs`, state-month by state-month



Black dots in (b) are binned means. The fitted slope is 0.12: a 1 m/s increase in TC wind shows up as about 0.12 m/s of ERA5-Land wind. In (a), the spike of points on the vertical axis is the 92.5% of state-months with no tropical cyclone — ERA5-Land wind there ranges over essentially its whole support.

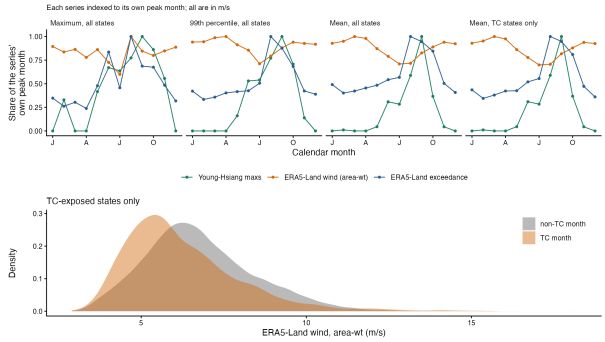
The same two series through time



How the two measures differ

- **Attribution.** `maxs` is LICRICE wind modelled from cyclone tracks, and is zero in 92.5% of state-months. ERA5-Land is the monthly maximum of daily maximum reanalysis wind, whatever the cause; its minimum is 2.9 m/s, mean 6.9.
- **Scale.** Regressing ERA5-Land on `maxs` within TC months gives a slope of 0.12.
- **Seasonality.** By monthly mean, raw ERA5-Land peaks in March and `maxs` in September. Across all state-months $r = 0.02$; within TC months $r = 0.49$.
- **Composition.** Of the 50 highest ERA5-Land wind months, 9 are TC months. The remainder are mostly winter months in Nebraska and Iowa, states with almost no tropical cyclones.

Seasonality, and separation of TC months



Top. `maxs` and the exceedance measure peak in cyclone season on every statistic; raw ERA5-Land does not — its 99th percentile peaks in April and its mean in March, both lowest in July–August. Its single *maximum* does fall in August, but that is one observation (Rhode Island, 1954) on a month whose 99th percentile is the second lowest of the year: 3 of the 100 windiest state-months are in August, and 31 of 100 in June–November against 50 if timing were flat. **Bottom.** Within TC states the densities largely overlap (6.23 vs 6.77 m/s), so the raw level does not separate TC months.

Swapping in ERA5-Land

The three ERA5-Land wind measures

All are state-month, in m/s. $w_{s\tau}$ is the ERA5-Land monthly maximum of daily maximum wind, averaged over state s.

1. **Area-weighted** $w_{s\tau}$ — the variable as delivered by the merge, grid cells averaged by area.
2. **Population-weighted** — same, averaged by LandScan population instead. Reported as a robustness; it behaves like (1) throughout.
3. **Exceedance**, written `w_exc`:

$$w^{\text{exc}}_{s\tau} \;=\; \max\Big(0,\;\; w_{s\tau} - q^{95}_{s,m(\tau)}\Big)$$

   where $q^{95}_{s,m}$ is the 95th percentile of ERA5-Land wind in state s and *calendar month* m, over 1950–2012.

Sample restrictions

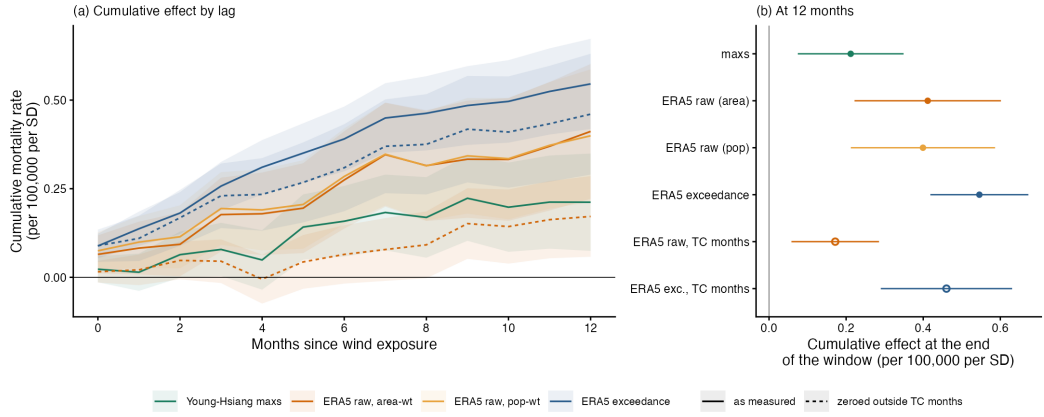
ERA5-Land wind starts in 1950. A 240-lag model needs 20 years of wind history before the first usable outcome, so the ERA5 regression begins in 1970, 20 years later than the replication.

Run different samples:

1. `maxs`**, full sample.** The replication from the previous section, estimated 1950–2012.
2. `maxs`**, ERA5-feasible sample.** The *same variable and the same specification*, re-estimated only on 1970–2012. Nothing about the measurement changes — only the sample length. This is the control.
3. **ERA5-Land area-weighted wind**, on that same 1970–2012 sample. This is the actual swap.

1-year window

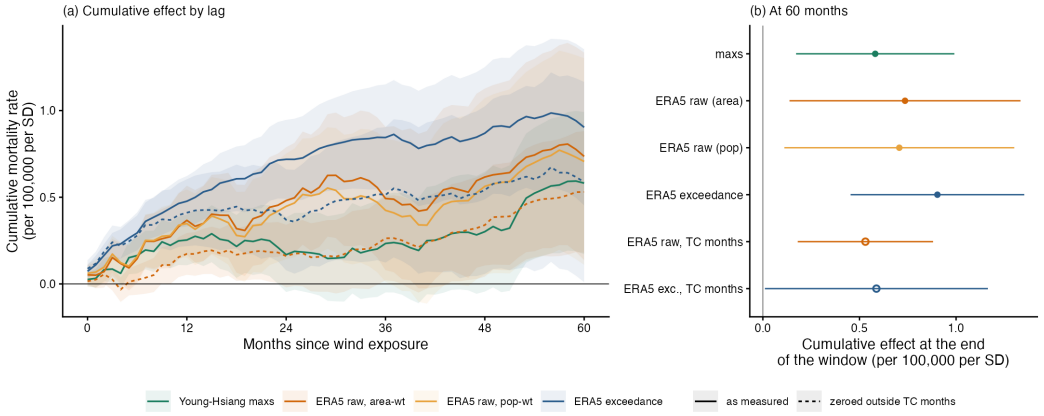
1951–2012, 26,784 obs



All series are signed at 12 months (t = 3.03, 4.25, 8.41; masked, 2.96 and 5.30). Masked to cyclone months, raw ERA5 falls from 0.412 to 0.172, below `maxs` (0.212), while exceedance changes little (0.546 → 0.460). Masked series use their *parent's* SD.

5-year window

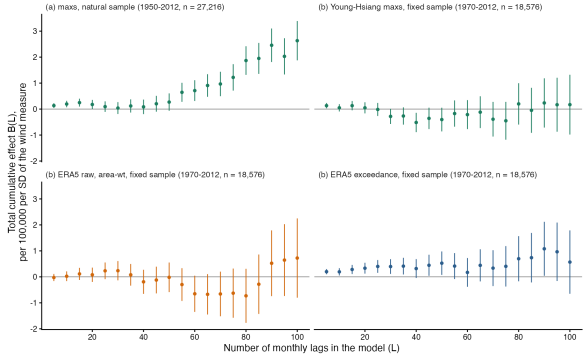
1955–2012 for every measure.



The masked series remain signed at this horizon ($t = 2.97$ raw, $t = 2.00$ exceedance).

Cumulative effect by lag window: first 100 lags

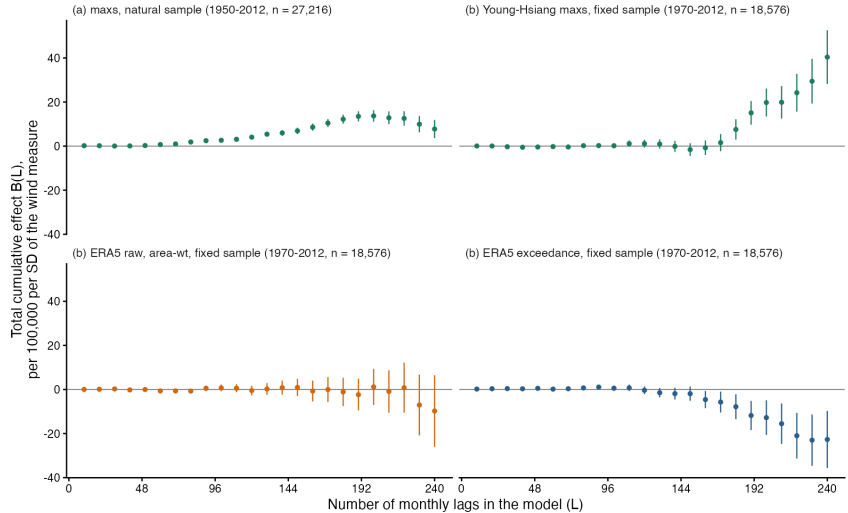
L = 5, 2×5, . . . , 100, on the identical samples, in SD units on a common vertical scale. Bars are 95% intervals.



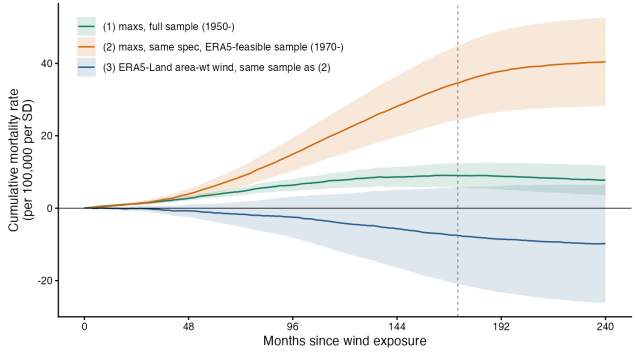
On its natural sample `maxs` is near zero to about 50 lags, then rises to 2.633 at 100 ($t = 6.82$). On the fixed 1970 sample the ERA5 series reach 0.724 (raw) and 0.569 (exceedance), with intervals widening in L.

Cumulative effect by lag window

L = 10, 2×10, . . . , 240 lags.



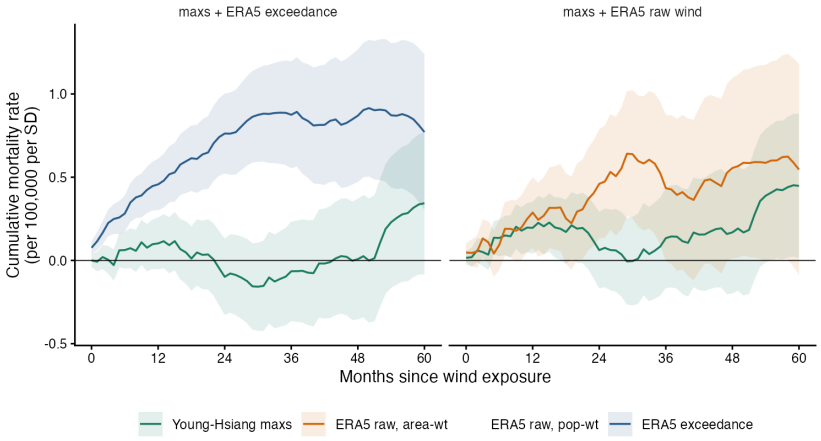
All lags included



(1) vs (2). Same variable and specification, shorter sample: the cumulative effect moves from 3.222 to 16.783, a factor of five with no change in measurement.

(2) vs (3). Identical sample (1970–2012, 18,576 observations) and identical specification; only the wind variable differs.

Horse race: both measures entered jointly



Entered jointly, the ERA5 exceedance measure remains signed ($t = 3.23$) while `maxs` attenuates ($t = 1.58$).

Why the ERA5 coefficients are larger

Four candidate explanations

1. **Scaling.** "Per SD" may not be equivalent for a sparse and an always-on series.
2. **Non-cyclone weather.** ERA5-Land may capture other damaging weather associated with mortality.
3. **Season or climate.** The association may be seasonal rather than storm-specific.
4. **Influential units.** A few states or years may drive it.

Everything that follows is on the 12-month window, 26,784 observations, produced by `03_era5_diagnostics.R`.

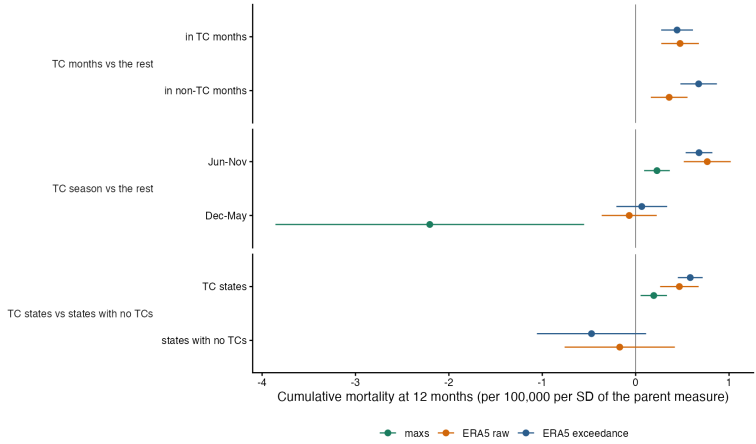
(1) Scaling

What "one standard deviation" means for each measure:

	Within-state SD (m/s)	% of months > 0	Mean when > 0 (m/s)	SD ÷ mean when > 0
`maxs` (Young–Hsiang)	2.41	7.5	7.29	0.33
ERA5 wind, area-weighted	1.15	100.0	6.88	0.17
ERA5 exceedance (state×month p95)	0.21	6.1	0.52	0.41

The last column is the size of a 1-SD move relative to a typical event in that variable.
It is of similar magnitude for `maxs` and the exceedance measure, so the per-SD comparison is not the source of the ordering.
Raw ERA5 wind is the outlier on sparsity: it is positive in every state-month, so its SD describes routine variation rather than events.

(2)–(3) Where the ERA5 signal sits



Each split is entered jointly, so the pair partitions the parent variable. Units are one SD *of the parent measure*. `maxs` (green) is absent from the first block because it defines that split (its non-TC half is identically zero), and from the no-TC states, which hold 9 TC months between them. Its Dec–May arm rests on 76 events, hence the wide interval.

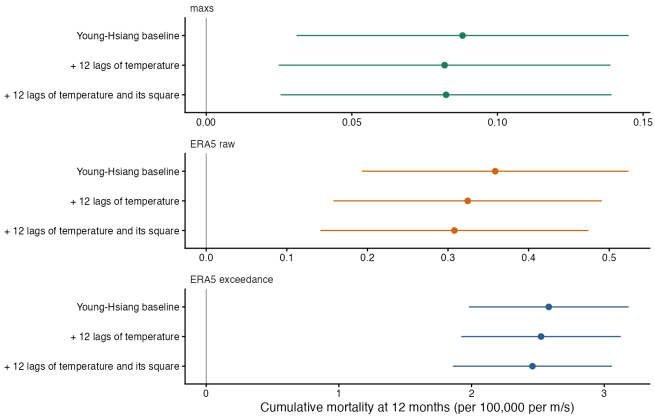
Diagnostic results

	Cumulative	(s.e.)	t
Reference: each measure on its own			
Young–Hsiang `maxs`	0.088	(0.029)	3.03
ERA5-Land raw	0.359	(0.084)	4.25
ERA5-Land exceedance	2.582	(0.307)	8.41
Split into TC and non-TC months, entered jointly			
ERA5-Land raw, in TC months	0.414	(0.089)	4.63
ERA5-Land raw, in non-TC months	0.313	(0.087)	3.58
ERA5-Land exceedance, in TC months	2.099	(0.411)	5.11
ERA5-Land exceedance, in non-TC months	3.192	(0.472)	6.76
Split by season, entered jointly			
ERA5-Land raw, Jun–Nov	0.668	(0.112)	5.97
ERA5-Land raw, Dec–May	-0.060	(0.131)	-0.46
ERA5-Land exceedance, Jun–Nov	3.210	(0.346)	9.29
ERA5-Land exceedance, Dec–May	0.310	(0.657)	0.47
Placebo: the 3 states with $<1\%$ TC months			
ERA5-Land raw	-0.149	(0.262)	-0.57
ERA5-Land exceedance	-2.237	(1.412)	-1.58

Cumulative effect at 12 months, deaths per 100,000 per m/s. Each "split" blocks enters both components in one regression, so they partition the parent variable.

Richer climate controls

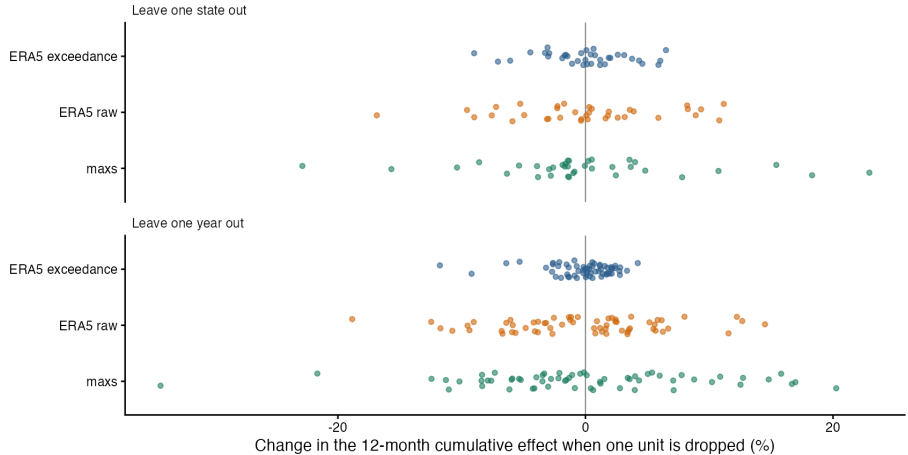
The baseline already carries Young and Hsiang's temperature control — a state-specific quadratic in *contemporaneous* monthly mean temperature.



On controlling for rainfall

We cannot, with this panel. Its only climate fields are `tavg` and the ERA5-Land wind columns — there is no precipitation variable in it. Could be added.

(4) Leave one state out, leave one year out



`maxs` is the fragile one: dropping a single unit moves it by up to 34%, and its t falls to 1.95 (dropping 1955; LA is the most influential state). The exceedance measure moves at most 12% and never drops below $t = 7.03$.

Summary of the diagnostics

1. The ERA5 result survives every placebo.
2. **But it is not measuring the same event set as `maxs`.** Roughly half of it sits in June–November months that Young and Hsiang code as having *no* tropical cyclone. Candidates: storms below their attribution threshold, near-misses whose wind field still reaches the state, and other severe convective weather in the same season.
3. **`maxs` is a narrower, noisier instrument.** It is zero in 92.5% of months, so it is identified off relatively few events — hence the jackknife fragility. ERA5-Land uses more of the panel.